Failure Secenario enlistment for KMRCL in line with letter no DMRC-134-12-RS3-DLP-3627 dtd: 12.01.2012

Failure Scenario	Scenario Code	System Code	Failure Class	Population	Alternate to withdrawal
Damage to any 1 or more TRCC	1.1.1	1.1	Collector Shoes	4 each side	NIL
Fuse Blown	1.1.2	1.1	Collector Shoes	4 each side	If power supply stabilisaes after resetting the fuse, withdrawal can be avoided.
TRCC Cable Damaged	1.1.3	1.1	Collector Shoes	8 each side	NIL
HSCB Tripping	1.2.1	1.2	HSCB & Earthing Switch	2 Cabs	If it trips again immediately after resetting from cab, train needs to be withdrawn.
PWM Generator Failure	1.3.1	1.3	Traction Inverter	2 Nos	NIL
Failiure one tranction motor.	1.4.1	1.4	Traction Motor	4 per Motor Bogie	Train can be operated in revenue with 50% degraded traction. Hence withdrawal is recommended when there is failure of more than 8 traction motors in a train.
Failiure to supply or control auxilary electrical equipment from either cab that disrupts train service.	2.1.1	2.1	Auxilliary Converter	4 Nos APS	Train to be withdrawan if more than one APS has failed in a train.
Battery under voltage during service	2.2.1	2.2	Battery Charger	4 Nos APS	NIL
Battery under voltage during service	2.3.1	2.3	Back-up Battery	4 Nos Battery	NIL
Loss of braking / Failure of brake control circuits in either cab	3.0.1	3.0	Air Supply & Frictionn Brake	1 set per train	NIL
Loss of MR / BP /PB pressure (below respective range)	3.0.2	3.0	Air Supply & Frictionn Brake	1 per Cab	NIL
Failure of train door control circuits at either cab	4.0.1	4.0	Door System & Controls	2 Cabs	NIL
Failiure of either cab door to open or close (Either physically or loss of electrical signal)	4.0.2	4.0	Door System & Controls	2 per Cab	NIL
Either cab partition door to open or close (Either physically or loss of electrical signal)	4.0.3	4.0	Door System & Controls	1 per Cab	NIL
Damage to any one or more door glass	4.1.1	4.1	Passenger Door	96	NIL
Failure of 2 train doors on the same side of a car even if the failed door can be isolated	4.1.2	4.1	Passenger Door	4 per car per side	NIL
Failiure of power supply circuit for saloon VAC or control circuits for saloon VAC in either driver's cab	5.0.1	5.0	VAC System & Controls	1 per Cab	NIL
Loss of ventilation or cooling to driver's cab	5.0.2	5.0	VAC System &	1 per Cab	NIL
Failure of more than one Saloon VAC in any one car	5.1.1	5.1	Saloon VAC	12 VACs	NIL

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Failiure of Public Announcement (Live PA broadcasting) during service.	6.0.1	6.0	Communication System	8 Saloon Speakers per car	There are 8 speakers per car with redundancy. Train should be withdrawan from revenue operation if all 8 speakers of any car stop functioning.
Failiure of Public Broadcsting (in TFT) during service.	6.0.2	6.0	Communication System	6 TFT per car	There are 6 TFT per car. Train should be withdrawan from revenue operation if all 6 TFTs of any car stop functioning.
Failure of CCTV HMI	6.0.3	6.0	Communication System	1 per cab	Even if the CCTV HMI fails, data will be recorded in DVR. No need to withdraw.
Public Address not possible from Cab to Saloon	6.0.4	6.0	Communication System	2 per cab	Train can be withdrawan if TO can not address to the passenger either from HMI/Microphone or from ACP
Passenger is not able to talk to train operator.	6.0.5	6.0	Communication System	4 per car	Train can be withdrawan if any 2 consecutive PEAU fail to communicate with TO
Failiure of coupler	7.0.1	7.0	Coupler & Draft Gear	5 Pairs	NIL
Poor Ride Quality due to loss of integrity of bogie / structure / wheelset / suspension / drive gear / bearings etc.	8.0.1	8.0	Bogies	12 Bogies	NIL
Failiure of head light high beam in leading direction.	9.1.1	9.1	Head Light	4 Nos	Withdraw if both the head light assembly fail in the leading DMC
TCMS CCU - if one CCU is offline or transmision error.	10.1.1	10.1	Control Unit	1 per Cab	If one CCU fails,the other CCU will continue to monitor/control the car signals and service operation
TCMS Control Node -One of the two CNs is offline or transmision error.	10.2.1	10.2	Local Unit (Communication	2 per car	If one CN fails,the other CN will continue to communicate with other sub systems and service
TCMS interface unit (RIO)-One of the two RIOs is offline or transmision error.	10.3.1	10.3	Car Interface Unit	2 per car	If one RIO fails,the other RIO will continue to monitor the car signals and service operation can be continued.
Video display Unit Failiure.	10.4.1	10.4	Display Controller	1 per Cab	NIL (In case of VDU failed,try to switch OFF and ON the control power MCB.If fault celars,service opertaion can be continued. Otherwise withdrawl)
Failure of structural integrity of any car	11.2.1	11.2	Carbody & Interior	6 Cars	If the car body damage is caused by any external factor, train will be withdrawan from revenue servises but it will counted as Non-Relevent failure.
Failure of driver's cab control which affects revenue operation	11.2.2	11.2	Carbody & Interior	2 Cabs	NIL
Degradation of driver's environment in rainy season	11.2.3	11.2	Carbody & Interior	2 Cabs	NIL

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Failure Scenario	Scenario Code	System Code	Failure Class	Population	Alternate to withdrawal
Exterior damage resulting into extention of kinematic envalope	11.3.1	11.3	Exterior - other than underframe	6 Cars	NIL
Breakage of one Windshield Glass.	11.3.2	11.3	Exterior - other than underframe	1 per cab	NIL
Damage to any one or more window glass	11.3.3	11.3	Exterior - other than	60 in a train	NIL
Loss of integrity of any equpment mounting.	11.4.1	11.4	Exterior -	6 Cars	NIL
Failiure of Gangway - Separation of Gangway	12.0.1	12.0	Gangway	5 Pairs	NIL